MASCAR: Multidomain Adaptive Spatial–Spectral Variable Compression Artifact Removal Network for Multispectral Remote Sensing Images

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***Abstract*— In remote sensing environments, image compression is essential to efficiently transmit and store high-resolution images due to the limited bandwidth and storage capacity. However, compression often leads to image quality degradation, requiring compression artifact removal technology in the postprocessing stage. Although deep neural networks have shown remarkable performance in image restoration, most existing methods have not adequately considered the compression conditions specific to remote sensing environments and have been evaluated primarily on synthetic datasets. To solve these issues, we propose a multidomain adaptive spatial–spectral variable compression artifact removal network (MASCAR) that effectively restores the earth surface details of compressed images in remote sensing environments. We introduce a multidomain local-patch collaborative learning strategy that extracts diverse features by decomposing the input local patch into different domains. In addition, we propose a detail focusing approach to direct the network's focus toward fine-texture detail restoration and ensure stable training of remote sensing images with significant deviations in pixel distribution of local patches. Furthermore, a detail enhancement approach is presented to enhance the details of the restored images. Moreover, we propose an incorporated compressed image quality adaptation mechanism to respond flexibly to unknown compression ratios in remote sensing environments. The performance of MASCAR applied with the proposed method is evaluated on synthetic and real-world remote sensing datasets. Experimental results demonstrate that the proposed method has better quantitative performance and visual quality than existing methods.**

***Index Terms*— Compression artifact removal, convolutional neural network (CNN), image restoration, remote sensing image (RSI).**

I. Introduction

REMOTE sensing via satellite or aircraft is a highly effective means of observing many phenomena on Earth, as it allows for data collection in dangerous or inaccessible areas. Remote sensing images (RSIs) analyze the Earth's surface and are used in various fields, including disaster monitoring, land usage classification, and weather observation [1], [2]. Generally, RSIs consist of multispectral (MS) and panchromatic (PAN) pairs, which occupy more storage space and bandwidth than natural scene images (NSIs). Therefore, an image compression method is essential to efficiently store and transmit high-resolution images in remote sensing environments with limited bandwidth and storage.

Several compression methods have been developed for efficiently compressing high-resolution RSIs, and among these, discrete cosine transform (DCT)-based compression methods [3] are popular due to their ease of hardware implementation, compatibility, and compression efficiency. Although these methods can achieve a high compression ratio, information is lost due to the quantization process. Therefore, various compression artifacts, such as ringing, aliasing, and blurring, inevitably occur in the decoded image. In particular, DCT-based compression methods quantize adjacent macroblocks into different coefficients, resulting in blocking artifacts due to boundary discontinuities. Therefore, compression artifacts can cause visual degradation and impair the performance of computer vision tasks.

Many studies have been proposed to mitigate compression artifacts. Generally, these methods are classified into two categories: model-based methods and learning-based methods. In the case of model-based methods, several approaches have been introduced to remove compression artifacts. These include filtering [4], [5], nonlocal similarity [6], [7], sparse coding [8], and so on. Typically, these approaches require manual design and fine-tuning using domain prior. Inspired by the great success of learning-based approaches in many image-processing applications, researchers have proposed learning-based methods for compression artifact removal. These methods aim to learn nonlinear mapping using training data. Wang et al. [9] and Chen et al. [10] introduced DCT- and wavelet-domain priors, respectively. Li et al. [11] trained a network with quantization tables as inputs and images to effectively remove local and global artifacts induced by compression. Jiang et al. [12] presented a network with compression quality predictor and controller for flexible compression artifact removal. Wang et al. [13] proposed a method of using contrastive quality representation learning to remove compression artifacts. More recently, Liang et al. [14]

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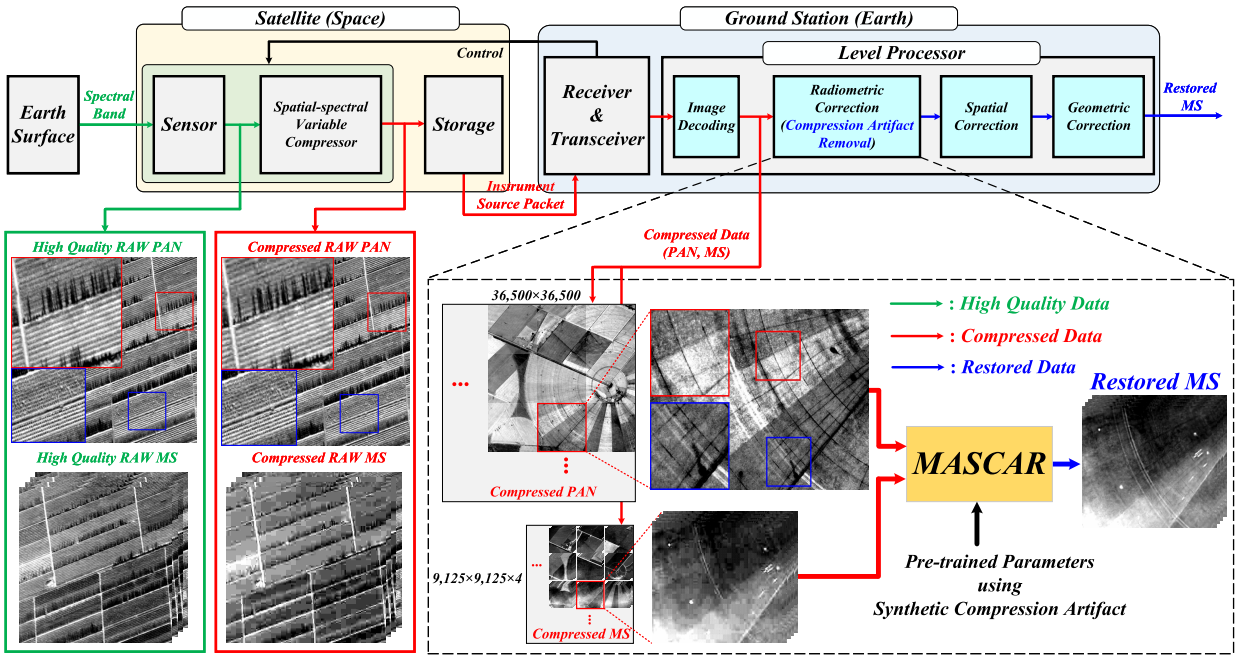


Fig. 1. Overall compressed RSI restoration pipeline.

presented transformer-based models for removing compression artifacts.

However, the methods above are unsuitable for remote sensing environments for the following reasons. First, unlike compression methods used in mobile and online services, the method for compressing RSIs applies the following techniques due to the unique properties of the remote sensing environments.

1) *Variable Quantization Table:* The quantization table changes according to the high-frequency content of small patches in the image.
2) *Bandwise Compression:* Due to limited transmission bandwidth and scanning, different compression ratios are applied to each sensor of the subband.
3) *Variable Bit Rate:* Due to memory and storage constraints in remote sensing environments, it is possible to be apply compression that exceeds the specified compression ratio.

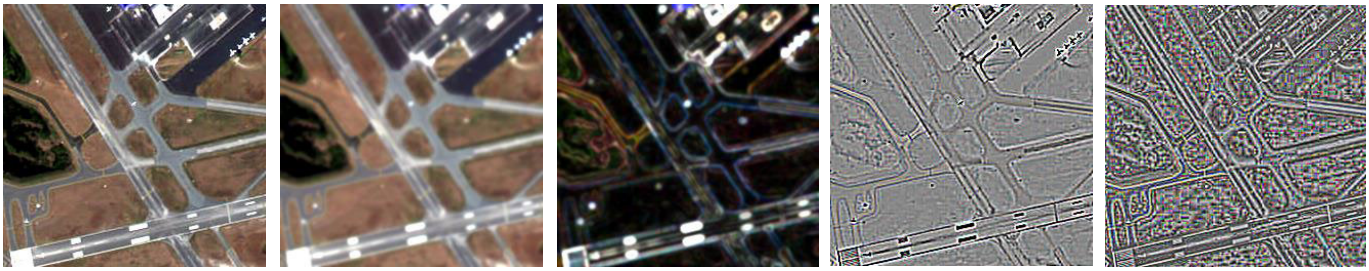
In summary, the spatial–spectral variable compression (SSVC) method is employed in remote sensing environments. Second, unlike NSIs where objects are centered, remote sensing environments' surface features are at high elevations, resulting in many flat areas and limited contextual information for image restoration. Finally, due to atmospheric conditions and ground surface geometric characteristics, careful model design is required for the stable training of RSIs, characterized by significant variations in pixel intensity distribution between small regions. To the best of our knowledge, there is no research specifically targeting the removal of lossy compression artifacts that consider SSVC with unknown compression ratios in remote sensing environments.

To address these problems, we propose a multidomain adaptive SSVC artifact removal network (MASCAR) that effectively restores compressed images in remote sensing environments. The overall image restoration pipeline with the proposed MASCAR is shown in Fig. 1. First, high-quality files obtained by the sensor are compressed and stored immediately after acquisition. Next, compressed raw files are transmitted to the ground station as instrument source packets and decoded into a digital image format. The decoded image still contains compression artifacts, and it is possible to obtain a restored image without compression artifacts by using the proposed MASCAR, which is part of the radiometric correction of the level processor.

We utilized a multidomain local-patch collaborative learning (MDLPCL) strategy for the proposed MASCAR training. As shown in Fig. 2, we first decompose the input image into four domains: mean, variance, normalized, and edge in the spatial domain. The spatially decomposed domain is a domain that expresses low- and high-frequency components by spatially decomposing the original image. This domain focuses on the intensity or properties of each pixel. Image processing in this domain emphasizes spatial characteristics and utilizes features such as texture, structure, and edge for image restoration. After that, the individually trained features from each domain are sequentially fused. The detailed image decomposition process is described in Section III. In addition, we designed a new loss function called multidomain collaborative criterion (MDCC) so that previous domain features can be used as effective priors for the subsequent domain training. Furthermore, we propose a detail focusing approach (DFA) and detail enhancement approach (DEA) to enhance the fine-texture detail of the restored image. Moreover, we propose an incorporated compressed image quality adaptation (ICIQA) mechanism to adaptively restore SSVC-applied RSIs. The main contributions of this article are summarized as follows.

1) Considering that various domains emphasize unique properties and compression artifacts differently, we propose the MDLPCL strategy to extract

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(a) (b) (c) (d) (e)

Fig. 2. Sample for pixel domain and spatially decomposed domains. (a) Pixel domain. (b) Mean domain. (c) Variance domain. (d) Edge domain. (e) Normalized domain.

nonredundant enriched features. In addition, we propose the MDCC to generate supervised domain outputs that operate as effective prior information for subsequent domain training.

2) We propose the DFA that directs the network's focus toward fine-texture detail restoration and effectively deals with RSIs with large deviations in pixel intensity distribution among small regions. In addition, we propose the DEA to enhance the spatial and edge details of the restored image.
3) To flexibly respond to SSVC with unknown compression ratios, we propose the ICIQA mechanism. The proposed ICIQA extracts discriminative features for compression artifacts through quality estimation and an adaptive reconstruction process. These features are used as pixel-wise attention maps during the refinement process to eliminate remaining artifacts and improve the image restoration quality.
4) Extensive experiments conducted on both synthetic and real-world datasets have shown that the proposed MASCAR outperforms other methods in terms of performance.

The remaining section of this article is organized as follows. Section II presents the related work. Section III details the proposed methods. Section IV reports datasets, experimental results, and ablation studies. Section V concludes this article.

II. Related Work

In this section, we briefly review the related work on learning-based compression artifact removal methods for NSIs and describe the recent RSI restoration methods.

A. Learning-Based Compression Artifact Removal for NSIs

In the past few years, learning-based approaches using deep neural networks have made significant progress and updated the state-of-the-art compression artifact removal performance every year. Dong et al. [15] were the first to introduce a learning-based approach to reduce compression artifacts. Wang et al. [9], Guo and Chao [16], and Zhang et al. [17] exploit DCT-domain priors to restore compressed images. Chen et al. [10] and Zhang et al. [18] proposed a wavelet-domain-based network to remove compression artifacts. Zheng et al. [19], Li et al. [11], and Ehrlich et al. [20] utilized chrominance components or quantization tables to remove color-domain compression artifacts. Jin et al. [21], [22] proposed networks that remove high- and low-frequency compression artifacts separately through structure-texture decomposition. Jiang et al. [12], Kim et al. [23], and Wang et al. [24] proposed an adaptive restoration network through compression quality estimation. Wang et al. [13] proposed a framework based on contrastive learning to remove artifacts from images with unknown compression ratios. Some work [25], [26] designed an advanced network architecture in which rich dependencies of deep features are modeled to remove compression artifacts. Liang et al. [14], Conde et al. [27], Zhang et al. [28], and Li et al. [29] presented a powerful vision transformer architecture for compressed artifact removal.

However, as mentioned in the introduction, it is a challenging task to apply existing works to RSIs directly. Existing works did not consider SSVC with unknown compression ratios in remote sensing environments. For example, the methods in [11], [16], [19], and [20] for requiring accurate quantization tables or DCT coefficients in the inference process are difficult to apply because the quantization tables vary according to the complexity of small regions. Furthermore, the methods in [9], [10], [15], [16], [17], and [29], which train a dedicated model for each compression ratio, are less flexible in environments with variable bit rate and variable quantization tables. In addition, in remote sensing environments where bandwise compression is applied, it is challenging to apply methods that remove only compression artifacts of luminance components [14], [21], [22], [25], [26] and [28] or methods that utilize chrominance components [11], [18], [19], [20]. Furthermore, most of these studies have been designed to remove compression artifacts in 8-bit natural RGB images, making them less applicable to RSIs with a significant variance in pixel intensity distribution. Finally, most of these works performed limited experiments on synthetic datasets, with little evaluation of real-world datasets.

B. RSI Restoration

RSIs are degraded by various sources (e.g., sensor noise, imaging system, and atmospheric conditions), making the interpretation of data very difficult. Therefore, many studies [1], [2] have been proposed to restore images from corrupted versions. Commonly, the RSI degradation can be

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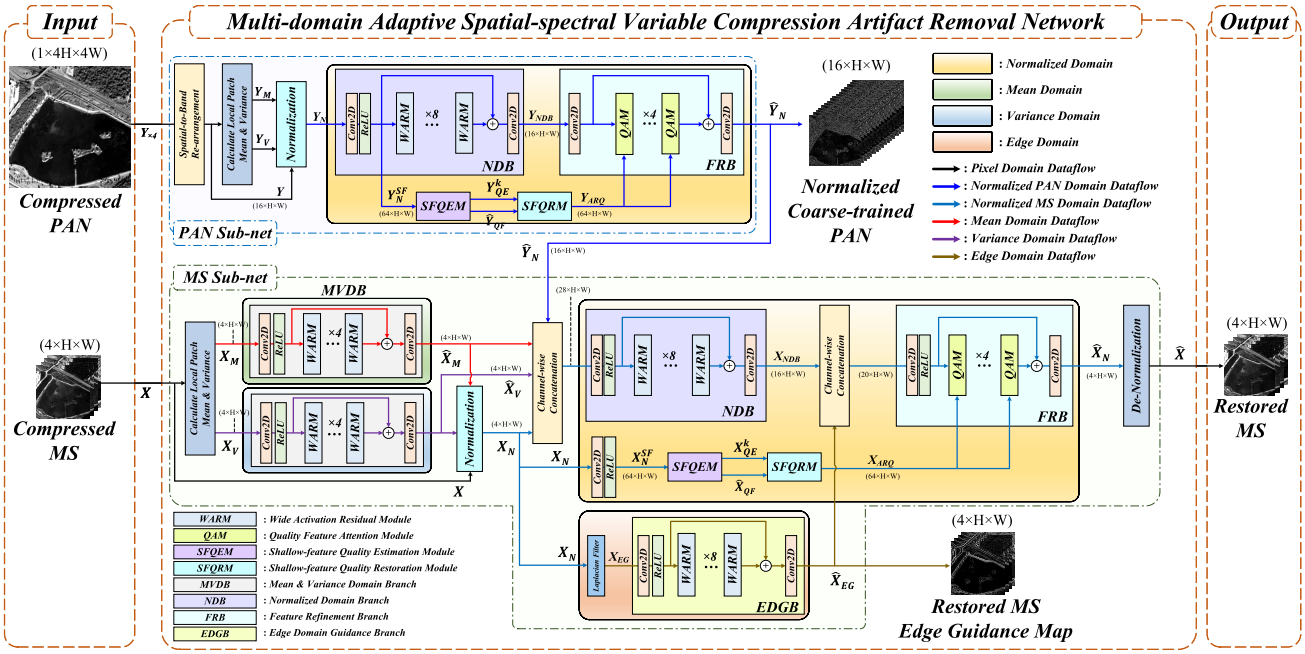


Fig. 3. Proposed MASCAR architecture.

modeled as

$$Y = K \times X + S + N \tag{1}$$

where X denotes the observed image, K denotes the multiplicative noise or blur kernel, S is the additive sparse noise modeled as a Laplace distribution, and N denotes the independent additive noise model. Many learning-based restoration methods have been proposed. Representative methods include denoising [30], [31], [32], [33], [34], despeckling [35], [36], [37], destriping [38], [39], and dehazing [40], [41], [42]. In addition, super-resolution [43], [44], [45] to obtain high-resolution images and spectral super-resolution [46], [47] methods to recover hyperspectral images from RGB images have also been proposed. Compared to existing image restoration models, compression artifact removal is the same ill-posed problem. However, denoising, which is actively studied in RSI restoration tasks, has the following differences. First, denoising can be modeled as a linear inverse problem. Still, compression artifact removal is a complex nonlinear inverse problem due to DCT-domain quantization and entropy coding applied in the compression process. Second, while noise is unintentional degradation due to the physical limitations of imaging sensors, imaging conditions, and environmental influences, compression artifacts are considered intentionally introduced degradation for the transmission and storage efficiency of RSIs.

However, few methods have been proposed to remove the compression artifacts of RSIs. Zhang et al. [48] proposed an end-to-end network to reduce compression artifacts. The authors designed a network to remove compression artifacts using high- and low-frequency information of images by combining summation and difference models. However, they removed only the compression artifacts of the luminance component of the RGB image and trained individual models for each compression ratio. In this article, we propose a novel network called MASCAR to address the challenges posed by the methods mentioned earlier and to effectively remove compression artifacts by considering SSVC with unknown compression ratios in remote sensing environments. The proposed network is a unified architecture that adaptively removes compression artifacts for a wide range of compression ratios by considering the discriminative features of the SSVC, including the variable quantization table, bandwise compression, and variable bit rate.

III. Methodology

In this section, we introduce the MASCAR for removing compression artifacts caused by SSVC with unknown compression ratios in remote sensing environments. First, we outline the network architecture of the proposed MASCAR. Then, we introduce the MDLPCL strategy. Subsequently, DFA and DEA are described. After that, we present the ICIQA mechanism. Finally, we explain the MDCC to train the proposed model.

A. Proposed Network Architecture Overview

The overall architecture of the network proposed in this article is presented in Fig. 3. The hierarchy of the proposed network is organized in the order of subnet, branch, and module from a top-down perspective. Modules combine to form branches, and branches gather to form a subnet. The proposed network comprises two subnetworks. The goal of the proposed network is to obtain the restored MS ($\widehat{X}$) given the compressed MS (X) and the compressed PAN ($Y_{\times 4}$)

$$\widehat{X} = G(X, F(Y_{\times 4})) \tag{2}$$

where $G(\cdot)$ denotes the MS subnet (MNet) and $F(\cdot)$ represents the PAN subnet (PNet).

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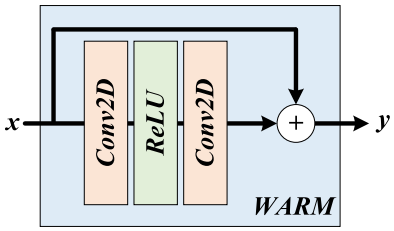


Fig. 4. WARM.

The PNet consists of the normalized domain branch (NDB), a feature refinement branch (FRB), a shallow feature quality estimation module (SFQEM), and a shallow feature quality reconstruction module (SFQRM). Specifically, PNet was designed to train solely in the normalized domain, taking the compressed PAN image as input and outputting a normalized coarse-trained PAN that is leveraged for spatial detail enhancement of the restored MS image

$$\widehat{Y}_N = F(Y_{\times 4}) \tag{3}$$

where $\widehat{Y}_N$ indicates the normalized coarse-trained PAN, which is used as a prior for normalized-domain training of MNet.

MNet is similar to PNet, but it incorporates the mean and variance domain branch (MVDB) and the edge domain guidance branch (EDGB). MNet is designed to take compressed MS and normalized coarse-trained PAN as input and output restored MS. The input MS image is decomposed into four domains, and training is performed on each domain branch. The features trained in the mean and variance domains are utilized as priors in the NDB, which trains the normalized domain, and the features trained in the edge domain are used as priors in the FRB. MNet and PNet utilize the SFQEM-SFQRM architecture to estimate compressed image quality using shallow features containing many compression artifacts and generate an adaptively reconstructed deep-quality feature map. In the FRB, this feature map is utilized as an attention map for selectively pixelwise feature refinement.

Inspired by Yu et al. [49], the proposed network employs the wide activation residual module (WARM) as the basic computational module for all domain branches, as shown in Fig. 4. The WARM can achieve much nonlinearity by expanding feature map channels before nonlinear activation

$$y = W(x) = c(\delta(c(x))) + x \tag{4}$$

where x and y represent the input and output of the WARM, respectively, and $W(\cdot)$ denotes the WARM. The dataflow of each domain branch is described as follows. First, shallow feature extraction is performed on the branch input

$$b_{\text{sf}} = \delta(c(b_{\text{in}})) \tag{5}$$

where $c(\cdot)$ denotes the convolutional layer and $\delta(\cdot)$ indicates the nonlinearity rectified linear unit (ReLU) function. In addition, b_{in} denotes the domain branch input, and b_{sf} represents the domain branch shallow features. Then, the extracted shallow features are mapped to deep features through the WARM streams, and the deep features are further processed using a convolutional layer to output the trained domain feature

$$b_{\text{out}} = c(W_k(\cdots W_1(b_{\text{sf}})\cdots) + b_{\text{sf}}), \quad k = 1, \ldots, N_B \tag{6}$$

where $W_k(\cdot)$ denotes to the kth WARM and b_{out} represents the domain branch output. Each branch was configured to train a mapping relationship for the residuals by inserting skip connections between domain shallow features and deep features. This way, the weights can be updated for small changes, increasing train stability and alleviating the vanishing gradient problem. In the proposed network, the number of shallow feature channels in the branch is 64, and the number of extended channels in WARM is 256. The channels of the convolutional layer located at the output of the branch are set as follows. EDGB, MVDB, and FRB are set to 4 for loss computation, and NDB is set to 16.

B. MDLPCL Strategy

In this article, we pay attention to the distinct characteristics captured in different domains. We propose the MDLPCL strategy that decomposes the input local patch into four domains, training each domain branch and sequentially fusing the previous domain output with the following domain input.

1) Motivation of Multidomain Decomposition: We decomposed the local patch from the pixel domain into four different domains, as depicted in Fig. 2. Each domain patch exhibits the unique properties of the image, such as color, structure, texture, and edge, being represented, revealing various artifacts that might not be identifiable in the pixel domain. We defined the unique properties prominently represented in each domain and the emphasized artifacts as domain properties and domain artifacts, respectively. The decomposed domain properties and artifacts are summarized as follows. First, color and structure properties are well represented in the mean domain, while color distortion artifacts are emphasized. Second, edge color and coarse texture properties are observed in the variance domain, as well as blocking and ringing artifacts. Third, object boundary, shape, and direction properties and blocking artifacts are identified in the edge domain. Finally, fine-texture properties blocking and aliasing artifacts are highlighted in the normalized domain.

The proposed MDLPCL strategy has three main characteristics. First, it extracts patchwise enriched local features. Training only in the pixel domain cannot effectively address artifacts of varying levels due to the redundant local features that occur in deeper networks. In contrast, the proposed MDLPCL strategy extracts nonredundant and enriched domain-specific features from various domains and performs powerful compression artifact removal. Second, it captures artifacts highlighted on a patchwise basis. By decomposing the pixel domain into multiple domains, diverse artifacts that are not easily identified in the pixel domain are highlighted as domain artifacts. Finally, the strategy utilizes progressive prior feature fusion, sequentially fusing the trained domain features as priors with the input of the subsequent domain. These priors guide removing compression artifacts in the subsequent domain training.

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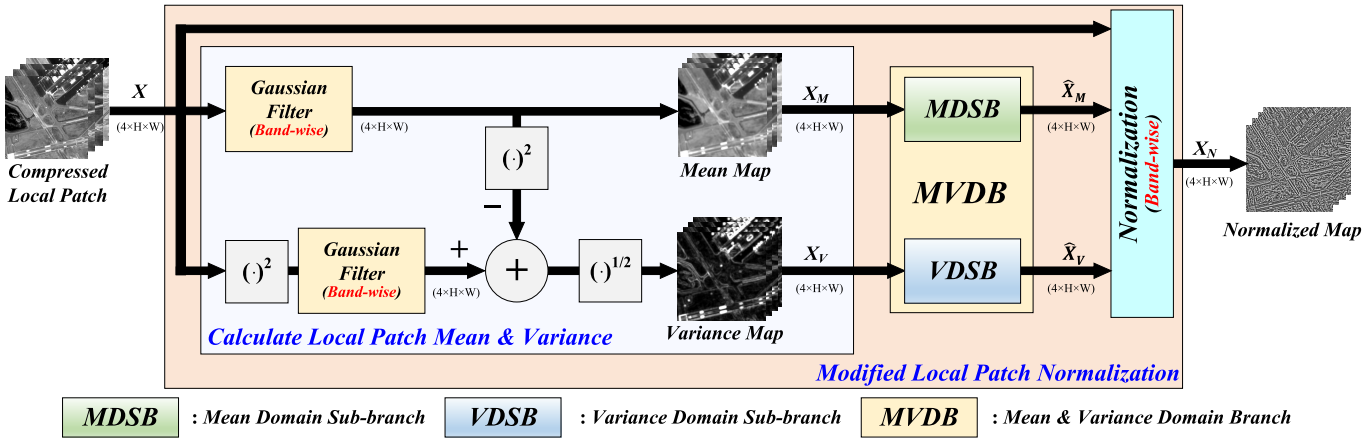


Fig. 5. Proposed modified local-patch normalization method.

2) Multidomain Decomposition: The process for decomposing the compressed MS image into four domains in MNet is described as follows. First, the pixel domain MS (X) is decomposed into the mean and variance domain using the Gaussian window for each channel, as illustrated in Fig. 5

$$X_M = K_G(X) \tag{7}$$

$$X_V = \sqrt{K_G(X^2) - K_G(X)} \tag{8}$$

where $K_G(\cdot)$ represents the Gaussian window. X_M and X_V represent mean and variance maps, respectively. Next, the mean and variance map are used to compute the normalized map (X_N) using the local-patch normalization method. However, as depicted in Fig 5, we used the modified local-patch normalization method to calculate the normalized map using the restored mean map ($\widehat{X}_M$) and the restored variance map ($\widehat{X}_V$) trained on the MVDB

$$\widehat{X}_M, \widehat{X}_V = G_{\text{MVDB}}(X_M, X_V) \tag{9}$$

$$X_N = (X - \widehat{X}_M)/\widehat{X}_V \tag{10}$$

where $G_{\text{MVDB}}(\cdot)$ denotes MVDB. The reasons for utilizing the restored mean and variance maps to compute the normalized map will be discussed in Section III-C1. Finally, the edge map (X_{EG}) for edge domain training is computed using a Laplacian filter

$$X_{\text{EG}} = \nabla(X_N) \tag{11}$$

where $\nabla(\cdot)$ denotes to the Laplacian filter. PNet is designed to be trained only in the normalized domain. Therefore, each channel's local Gaussian window is applied to compute the mean map (Y_M) and the variance map (Y_V), followed by normalized map calculation using local-patch normalization

$$Y_N = (Y - Y_M)/Y_V \tag{12}$$

where Y represents the pixel domain PAN and Y_N denotes the normalized map of PNet.

3) Progressive Prior Feature Fusion: After removing domain artifacts from the previous domain branch, decomposed domain features are fused with the following domain input. The normalized coarse-trained PAN ($\widehat{Y}_N$) trained on the normalized domain of PNet, the restored mean map ($\widehat{X}_M$), and the restored variance map ($\widehat{X}_V$) are utilized as priors for training the normalized domain of MNet. These highly spatial–spectral correlated priors are utilized as effective guidelines for removing compression artifacts. The three priors are channelwise concatenated to the normalized map, which is the input to the NDB of the MNet

$$X_{\text{NDB}} = G_{\text{NDB}}([X_N, \widehat{Y}_N, \widehat{X}_M, \widehat{X}_V]) \tag{13}$$

where $G_{\text{NDB}}(\cdot)$ denotes the NDB of the MNet, X_{NDB} represents the output of the NDB, and $[\cdot]$ means channelwise concatenation. The edge guidance map ($\widehat{X}_{\text{EG}}$) trained on the EDGB of the MNet is channelwise concatenated with X_{NDB} and input to the FRB of the MNet

$$\widehat{X}_N = G_{\text{FRB}}([X_{\text{NDB}}, \widehat{X}_{\text{EG}}], X_{\text{ARQ}}) \tag{14}$$

where $G_{\text{FRB}}(\cdot)$ denotes the FRB of MNet and $\widehat{X}_N$ denotes the output of the FRB. In addition, X_{ARQ} represents the adaptively reconstructed deep-quality feature map, the output of the SFQRM, which will be described in Section III-D. This feature map plays a crucial role in improving the image restoration quality in the feature refinement process.

C. Detail Focusing and Detail Enhancement Approaches

1) Detail Focusing Approach: In this article, we propose a DFA to effectively handle RSIs and allow the network to focus on texture detail restoration. Generally, uniform normalization, which uses precalculated dataset statistics, or global normalization, which utilizes individual input statistics, stabilizes network training. However, these methods cause training instability when applied to RSIs, which have a wide dynamic range and significant deviations of pixel intensity distributions in small regions depending on the geographical characteristics of the surface.

To address these issues, we introduce local-patch normalization that uses patchwise local window statistics described in Section III-B2. This normalization method calculates the mean and variance maps using the Gaussian window for each channel to obtain a normalized map that follows the Gaussian distribution in the input patch. This approach ensures stable and generalized normalization for RSIs within the network,

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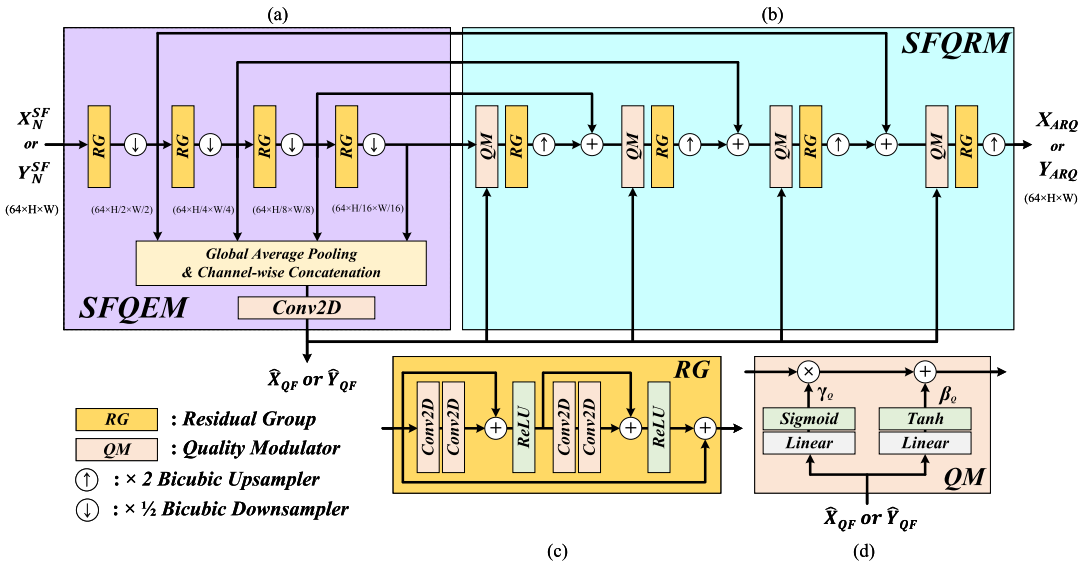


Fig. 6. Proposed SFQEM-SFQRM architecture. (a) SFQEM. (b) SFQRM. (c) RG. (d) QM.

alleviating color shifting by preserving color information in the mean and variance maps. We found that a 9×9 Gaussian window with 1.5 sigma provides good generalization performance and applied it to the proposed network.

However, in the denormalization process, domain artifacts in the mean and variance maps are reflected in the final output. To solve this problem, the domain artifacts of the mean and variance map were removed from the MVDB shown in Figs. 3 and 5. This branch consists of the mean domain subbranch and the variance domain subbranch to remove the domain artifacts of the mean and variance map, respectively. Therefore, the final output does not contain compression artifacts because the denormalization is performed using the restored mean and variance maps with domain artifacts removed.

The normalized map shown in Fig. 2(e) shows that the fine-texture detail property is well represented, and the blocking and aliasing artifacts are highlighted because the redundant information of neighboring pixels is removed. Therefore, it can focus on restoring texture detail more than pixel domain training, and it can effectively remove compression artifacts because it comes with highly spatial–spectral correlated priors.

2) Detail Enhancement Approach: Since many high-frequency components are removed by the quantization process during image compression, the restored image often appears blurry. To address this issue, we propose a DEA. The proposed DEA consists of early coarse-normalized PAN fusion and late MS edge fusion techniques that aim to improve the spatial and edge details of the restored MS, respectively.

First, we describe the early coarse-normalized PAN fusion technique [50], [51]. The PAN image covers a wide range of wavelengths, and its spatial resolution is four times bigger than the MS image. Therefore, we designed the PNet to incorporate the enriched spatial details of the PAN into the restored MS. Two methods were applied for the effective fusion of the output of the PNet with the features of the MNet. First, we matched the spatial resolution of PNet to that of the MS image using the space-to-band rearrangement layer [52], which was used to increase the number of channels in the PAN image by a factor of 16 while reducing the spatial resolution by a quarter

$$Y = F_{\mathrm{S2BRA}}(Y_{\times 4}) \tag{15}$$

where $F_{\mathrm{S2BRA}}(\cdot)$ denotes the space-to-band rearrangement layer. When H and W are the spatial resolution of the MS image, the dimensions of $Y_{\times 4}$ are $1 \times 4\mathrm{H} \times 4\mathrm{W}$, and the dimensions of Y are $16 \times \mathrm{H} \times \mathrm{W}$. Unlike other downsampling methods, it reduces the resolution without modifying the original pixels, ensuring no loss of information. Moreover, it alleviates misalignment issues from spatial resolution disparities during the MS and PAN fusion process.

Second, we configured the PNet to be trained on the normalized domain for stable fusion of MS and PAN. Since the normalized domain has almost no color and structure properties and only texture properties are emphasized, the stable fusion between MS and PAN features with low spectral correlation is possible. The dataflow of PNet is given as follows. First, shallow feature extraction is performed for normalized domain learning in the PNet

$$Y_N^{\mathrm{SF}} = \delta(c(Y_N)) \tag{16}$$

where Y_N^{SF} denotes the normalized shallow feature of PNet. The normalized shallow feature of the PNet is used as input for texture detail restoration in the NDB

$$Y_{\mathrm{NDB}} = F_{\mathrm{NDB}}\left(Y_N^{\mathrm{SF}}\right) \tag{17}$$

where $F_{\mathrm{NDB}}(\cdot)$ is the NDB of PNet. The FRB performs pixelwise feature refinement by utilizing the output of the NDB (Y_{NDB}) and the adaptively reconstructed deep-quality feature map (Y_{ARQ}) calculated by the SFQEM and the SFQRM as attention maps

$$\widehat{Y}_N = F_{\mathrm{FRB}}\left(Y_{\mathrm{NDB}}, Y_{\mathrm{ARQ}}\right) \tag{18}$$

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where $F_{\mathrm{FRB}}(\cdot)$ denotes the FRB of PNet, and $\widehat{Y}_N$ represents the normalized coarse-trained PAN that is the output of PNet. The FRB of the MNet is similar to PNet, and a detailed description of the FRB is described in Section III-D.

Next, we designed the EDGB to enhance edge details such as diagonal lines and the shape and orientation of objects in the restored image. The restored edge guidance map is channelwise concatenated with the output of the NDB and utilized in the feature refinement process to enhance the edge details of the final MS output. The architecture of the EDGB is the same as NDB, except for including a Laplacian filter at the input stage. We applied the 5×5 Laplacian window to the network after confirming that it extracts appropriate edges and does not degrade the quality of the restored image.

D. Incorporated Compressed Image Quality Adaptation Mechanism

In this article, we propose an ICIQA mechanism to respond flexibly to SSVC with unknown compression ratios in remote sensing environments. The proposed ICIQA mechanism consists of three steps. First, in the quality estimation step, the image quality factors (QFs), a quality measure of DCT-based compression, are estimated by learning various compression quality levels from shallow features with many compression artifacts. Next, in the quality reconstruction step, an adaptively reconstructed deep-quality feature map is generated using the estimated QFs and the various compression representations. Finally, in the feature refinement step, pixelwise feature refinement is conducted using the adaptively reconstructed deep-quality feature map as attention maps to improve restoration quality. As shown in Fig. 6(a) and (b), the proposed SFQEM-SFQRM uses an encoder–decoder architecture. In this architecture, when a pair with the same spatial resolution of feature maps is regarded as one stage, the proposed SFQEM-SFQRM consists of four stages. The basic computation module used in SFQEM-SFQRM is the residual group (RG), which can be seen in Fig. 6(c).

The SFQEM utilizes four RG and downsampler pairs, extracting multiscale features from the four-stage encoder to estimate compressed image quality

$$X_{\mathrm{QE}}^{k} = \mathrm{BD}\left(\mathrm{RG}\left(X_{\mathrm{QE}}^{k-1}\right)\right) \tag{19}$$

where $X_{\mathrm{QE},k}$ denotes the output of the kth stage in SFQEM, $\mathrm{BD}(\cdot)$ represents the bicubic downsampler, and $\mathrm{RG}(\cdot)$ denotes the RG. The multiscale features are transformed into 1-D vectors using global average pooling (GAP) and then channelwise concatenated to form a unified encoded feature. This feature is adjusted to four channels to estimate bandwise QFs using 1×1 convolutional layer

$$\widehat{X}_{\mathrm{QF}} = c\left(\left[\mathrm{GAP}(X_{\mathrm{QE}}^{1}), \ldots, \mathrm{GAP}(X_{\mathrm{QE}}^{4})\right]\right) \tag{20}$$

where $\widehat{X}_{\mathrm{QF}}$ denotes the QFs of each estimated subband and has the dimensions 4×1. The SFQRM utilizes four RG, quality modulator (QM), and upsampler pairs, generating the adaptively reconstructed deep-quality feature map using encoded compression quality features and estimated bandwise QF

$$X_{\mathrm{ARQ}} = \mathrm{SFQRM}\left(X_{\mathrm{QE}}^{1}, \ldots, X_{\mathrm{QE}}^{4}, \widehat{X}_{\mathrm{QF}}\right) \tag{21}$$

where X_{ARQ} denotes the adaptively reconstructed deep-quality feature map and $\mathrm{SFQRM}(\cdot)$ represents an SFQRM in the MNet. The design of the proposed QM is inspired by the spatial feature transformation method presented in [53] and [54]. This transform adaptively modulates output features by spatially applying affine transformation parameters dynamically trained from external information. In this study, we designed the QM specialized for compression quality feature reconstruction by utilizing the estimated QFs as external priors. As shown in Fig. 6(d), the QM is trained to generate elementwise quality modulation pairs (γ_Q, β_Q), and the affine transformation in the QM is specified as follows:

$$\mathrm{QM}\left(X_{\mathrm{QR}}^{k}, \widehat{X}_{\mathrm{QF}}\right) = \gamma_Q \cdot X_{\mathrm{QR}}^{k} + \beta_Q \tag{22}$$

where X_{QR}^{k} denotes the QM input feature map of the kth stage. Through a specialized QM, it can adaptively respond to various compression quality levels. The number of internal channels for SFQRM-SQFEM is 64. In addition, γ_Q and β_Q are set to 64 output features for the affine transformation. The SFQEM-SFQRM of PNet has the same architecture as MNet, and the output channel of the convolutional layer is adjusted to one to estimate the QF of PAN.

We utilized the adaptively reconstructed deep-quality feature map generated from SFQEM-SFQRM as attention maps for pixelwise feature refinement. Since this feature map is adaptively modulated and reconstructed from the estimated QFs and learned with different compression quality levels, it is an ideal guide for feature refinement. The quality attention map, dynamically selected by the attention module, provides information about where the network should focus during the image restoration. Therefore, the quality attention maps improve the restoration quality by removing the remaining compression artifacts through pixelwise feature refinement in the FRB. As shown in Fig. 7(a), the quality attention module (QAM) is the basic computational module of the FRB, which consists of the WARMs, a spatial-attention module (SAM), and a channel-attention module (CAM). The QAM works as follows. First, the adaptively reconstructed deep-quality feature map is transformed into attention maps using CAM and SAM. As shown in Fig. 7(b), the CAM uses the GAP in the input feature map to reduce the spatial resolution to 1×1. It utilizes internal channel relationships to generate a 1-D attention map ($C \times 1 \times 1$)

$$X_{Q}^{\mathrm{CA}} = F_{Q}^{\mathrm{CAM}}(X_{\mathrm{ARQ}}) = \sigma(c(\gamma(c(\mathrm{GAP}(X_{\mathrm{ARQ}})))))) \tag{23}$$

where X_{Q}^{CA} represents the channelwise quality attention map and $\sigma(\cdot)$ denotes the sigmoid function. $F_{Q}^{\mathrm{CAM}}(\cdot)$ denotes the CAM of MNet, and the number of channels is set to four. As shown in Fig. 7(c), the SAM reduces the input feature map into a single channel to generate a 2-D attention map ($1 \times \mathrm{H} \times \mathrm{W}$)

$$X_{Q}^{\mathrm{SA}} = F_{Q}^{\mathrm{SAM}}(X_{\mathrm{ARQ}}) = \sigma(c(\gamma(c(X_{\mathrm{ARQ}}))))) \tag{24}$$

where X_{Q}^{SA} denotes the spatial quality attention map and $F_{Q}^{\mathrm{SAM}}(\cdot)$ represents the SAM, and the number of channels

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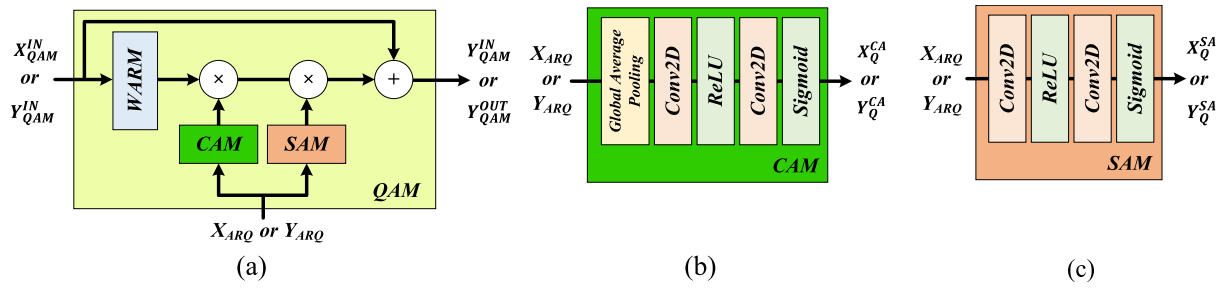


Fig. 7. Various modules used in the FRB. (a) QAM. (b) CAM. (c) SAM.

is set to four. Next, pixelwise multiplication is sequentially performed on the two attention maps for the WARM's output

$$X_{\mathrm{QAM}}^{\mathrm{OUT}} = W\left(X_{\mathrm{QAM}}^{\mathrm{IN}}\right) \otimes X_{Q}^{\mathrm{CA}} \otimes X_{Q}^{\mathrm{SA}} + X_{\mathrm{QAM}}^{\mathrm{IN}} \tag{25}$$

where $\otimes$ denotes pixelwise multiplication, and $X_{\mathrm{QAM}}^{\mathrm{IN}}$ and $X_{\mathrm{QAM}}^{\mathrm{OUT}}$ are the input and output feature map of QAM, respectively. Using two attention maps, feature refinement can be performed on all pixels of the 3-D tensor, effectively improving the restoration quality.

The proposed SFQEM-SFQRM adopts the encoder–decoder architecture that is similar to the decoupler-reconstructor introduced in the FBCNN [12]. However, there are several differences in the details. First, there is a difference in utilizing the encoder–decoder. In the FBCNN, the encoder–decoder architecture is used as an end-to-end network to reconstruct the image. However, in the proposed method, the encoder–decoder architecture is used as an auxiliary branch. The adaptively reconstructed deep-quality feature map serves as the output and generates attention maps for pixelwise feature refinement. Next, there is a difference in how the image quality is estimated. While the FBCNN uses features from the last stage of the encoder to estimate a QF per image, the proposed method utilizes multiscale features used in encoding to estimate bandwise QFs. Finally, the FBCNN aims to reconstruct the image with the desired QF, considering the user's preference. In contrast, the proposed SFQEM-SFQRM method seeks to provide optimal guidance for effectively removing compression artifacts. This approach is designed to respond to the challenges of handling RSIs affected by SSVC with unknown compression ratios.

E. Multidomain Collaborative Criterion

In this article, we propose a novel loss function called the MDCC for collaborative learning of multidomain branches. Supervised domain outputs guided by the proposed MDCC serve as effective priors for subsequent domain training. All the criteria that comprise the MDCC are L1, and the losses for individual bands are calculated and added together. As shown in Fig. 8, the MDCC is defined as follows:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_R + \epsilon_{\mathrm{NP}} \cdot \mathcal{L}_{\mathrm{NP}} + \epsilon_E \cdot \mathcal{L}_E + \epsilon_{\mathrm{EGM}} \cdot \mathcal{L}_{\mathrm{EGM}} \\ + \epsilon_{\mathrm{MV}} \cdot \mathcal{L}_{\mathrm{MV}} + \epsilon_Q \cdot \mathcal{L}_Q\end{aligned} \tag{26}$$

where $\mathcal{L}_R$, $\mathcal{L}_{\mathrm{NP}}$, $\mathcal{L}_E$, $\mathcal{L}_{\mathrm{EGM}}$, $\mathcal{L}_{\mathrm{MV}}$, and $\mathcal{L}_Q$ are the MS restoration loss, normalized PAN loss, edge loss, edge guidance map loss, mean and variance losses, and quality loss, respectively.

First, the restoration loss ($\mathcal{L}_R$) is the primary loss function for image restoration in the proposed network, corresponding to the red line in Fig. 8. It is used to minimize the loss function between the original MS ($\bar{X}$) and the restored MS ($\hat{X}$).

Second, the normalized PAN loss ($\mathcal{L}_{\mathrm{NP}}$) is the loss function for PNet training, corresponding to the blue line in Fig. 8. The normalized PAN loss minimizes the difference between the normalized coarse-trained PAN ($\hat{Y}_N$), the output of the PNet, and the normalized original PAN ($\bar{Y}_N$). Using the normalized PAN loss, the normalized coarse-trained PAN can be used as a prior for the normalized domain training of the MNet while fusing spatial details into the MS.

Third, edge loss ($\mathcal{L}_E$) is a loss function that directly enhances the edge detail of the restored image, which corresponds to the green line in Fig. 8. The edge loss consists of two sublosses: the downscaled PAN edge loss ($\mathcal{L}_{\mathrm{DPE}}$) and the MS edge loss ($\mathcal{L}_{\mathrm{ME}}$). The input to both sublosses is the restored MS edge ($\hat{X}_E$). The target of the downscaled PAN edge loss is the PAN ($\bar{Y}_{\mathrm{DE}}$) with bicubic downsampling and a first-order derivative filter applied sequentially. The target of the MS edge loss is the Laplacian edge of the original MS ($\bar{X}_E$). The overall edge loss is defined as

$$\mathcal{L}_E = \mathcal{L}_{\mathrm{ME}} + \mathcal{L}_{\mathrm{DPE}}. \tag{27}$$

Fourth, the edge guidance map loss ($\mathcal{L}_{\mathrm{EGM}}$) is the loss function for EDGB training in MNet, which corresponds to the orange line in Fig. 8. The edge guidance map loss consists of two sublosses: the downscaled PAN edge guidance map loss ($\mathcal{L}_{\mathrm{DPEGM}}$) and the MS edge guidance map loss (L_{MEGM}). The edge guidance map loss uses the same targets as the edge loss; only the input is changed in the restored MS edge guidance map ($\hat{X}_{\mathrm{EG}}$). The overall edge guidance map loss is defined as

$$\mathcal{L}_{\mathrm{EGM}} = \mathcal{L}_{\mathrm{MEGM}} + \mathcal{L}_{\mathrm{DPEGN}}. \tag{28}$$

Fifth, the mean–variance loss ($\mathcal{L}_{\mathrm{MV}}$) is the loss function for MVDB training, corresponding to the purple line in Fig. 8. The mean–variance loss comprises two sublosses: the mean loss ($\mathcal{L}_M$) and the variance loss ($\mathcal{L}_V$). The restored MS mean and variance maps minimize the loss from the original MS mean and variance maps, respectively. The overall mean–variance loss is defined as

$$\mathcal{L}_{\mathrm{MV}} = \mathcal{L}_M + \mathcal{L}_V. \tag{29}$$

Finally, the quality loss is composed of two sublosses, the PAN quality loss (L_{PANQ}) and the MS quality loss (L_{MSQ}),

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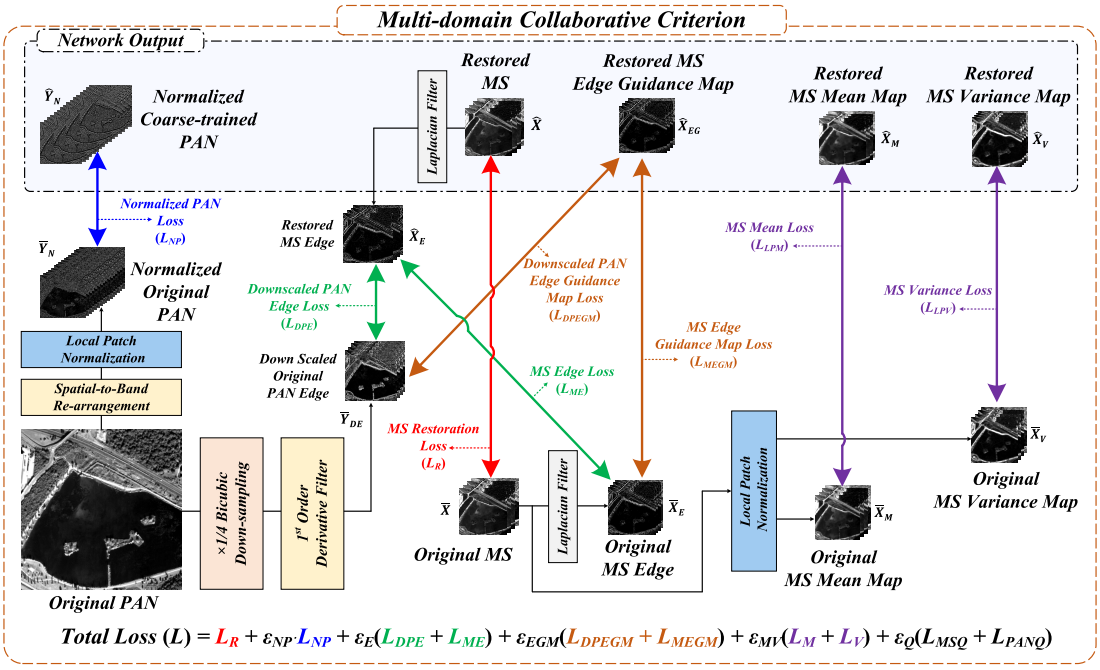


Fig. 8. Proposed MDCC.

respectively,

$$\mathcal{L}_{\mathrm{MSQ}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{c=1}^{4} \|\bar{X}_{\mathrm{QF}}[b,c] - \hat{X}_{\mathrm{QF}}[b,c]\|_1 \tag{30}$$

$$\mathcal{L}_{\mathrm{PANQ}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{c=1}^{16} \|\bar{Y}_{\mathrm{QF}}[b,c] - \hat{Y}_{\mathrm{QF}}[b,c]\|_1 \tag{31}$$

where B denotes the batch size and c is the index of each band. In addition, $\hat{X}_{\mathrm{QF}}$ and $\hat{Y}_{\mathrm{QF}}$ denote the estimated QFs of the MS and PAN, and $\bar{X}_{\mathrm{QF}}$ and $\bar{Y}_{\mathrm{QF}}$ are the ground-truth QFs of MS and PAN images, respectively. Therefore, the overall quality loss is defined as

$$\mathcal{L}_Q = \mathcal{L}_{\mathrm{MSQ}} + \mathcal{L}_{\mathrm{PANQ}}. \tag{32}$$

We group loss functions that play a similar role and give them weight. The loss function weights ($\epsilon_{\mathrm{NP}}, \epsilon_E, \epsilon_{\mathrm{EGM}}, \epsilon_{\mathrm{MV}}$, and ϵ_Q) are empirically set to 0.01, 0.5, 0.5, 0.001, and 0.01, respectively.

IV. Experiment Results

In this section, we introduce details about the experimental setting and dataset for training the proposed MASCAR. After, we provide a detailed discussion of extensive experimental results, evaluating the proposed MASCAR against several state-of-the-art methods, focusing on both quantitative performance metrics and qualitative visual results.

A. Dataset

The Korea Aerospace Research Institute (KARI) dataset was utilized to validate the proposed network in this article. The dataset has a ground sampling distance of 0.5 m for the MS and 2.0 m for the PAN. The digital number of the images is 12 bit. The dataset is divided into two subdatasets: one with very low compression ratios and almost no compression artifacts (KARI Nocomp dataset) and the other applied SSVC with unknown compression ratios (KARI Blind Comp dataset). The KARI Nocomp dataset consists of 20 MS and PAN pairs. For training and validation of the proposed network, the KARI Nocomp dataset was cropped, and bandwise DCT-based compression was applied to generate a synthetic dataset (KARI Synthetic Comp dataset). The size of MS and PAN in the KARI synthetic Comp dataset is 256×256 and 1024×1024, respectively, and consists of 7779 pairs. A total of 7636 pairs were used as training data, and the remaining were used for validation. In addition, the KARI Blind Comp dataset consists of five pairs of $36\,500 \times 36\,500$ PAN and 9125×9125 MS images. Because the entire large-scale image could not be loaded into the GPU memory, the original image was cropped to a size of 256×256, and 2025 acquired patches were used for inference. Result patches are merged back to their original size.

B. Experiment Setting

All the experiments were conducted on a workstation with Intel[1] Xeon[1] Gold 6226R CPU @ 2.90 GHz, NVIDIA GTX 3090 GPU, and 256-GB memory. The proposed MASCAR was trained with momentum $\beta_1 = 0.9$, $\beta_2 = 0.999$, and ϵ = 1e-7 using the Adam algorithm as a gradient descent optimization method, and a batch size of 16 was used. The initial learning rate was set to 5e-5 and halved every 1e5 iteration until a final learning rate of 7.8125e-7 was reached. In addition, the QF was set to [10, 40], and randomly cropped images with 96×96 patches were used as training data.

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TABLE I

NSI NETWORK GROUP COMPARISON RESULTS OF AVERAGE PSNR, SSIM, PSNR-B, SAM, RASE, AND ERGAS ON THE KARI SYNTHETIC COMP DATASET (THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY)

QF	Metric	Methods												
		MS	NSSRC [6]	HSSE [7]	SwinIR [14]	FBCNN [12]	COLA-Net [26]	MIRNetv2 [55]	Restormer [56]	Swin2SR [27]	ART [28]	GRL [29]	MASCAR (Ours)	MASCAR+ (Ours)
10	PSNR↑	46.51	47.75	47.84	48.80	48.60	49.19	48.86	49.12	49.20	49.15	49.24	49.34	52.52
	SSIM↑	0.9826	0.9901	0.9902	0.9901	0.9901	0.9909	0.9902	0.9908	0.9909	0.9908	0.9910	0.9912	0.9960
	PSNR-B↑	43.95	46.45	46.62	48.65	48.27	49.19	48.68	48.96	48.88	49.10	49.00	49.21	52.15
	SAM↓	0.0275	0.0233	0.0230	0.0203	0.0209	0.0194	0.0205	0.0198	0.0196	0.0195	0.0193	0.0191	0.0136
	RASE↓	168.527	144.991	143.374	127.669	131.091	122.723	129.412	125.233	123.786	123.254	121.906	120.678	85.1264
	ERGAS↓	337.055	289.982	286.748	255.337	262.181	245.447	258.823	250.465	247.572	246.509	243.811	241.356	170.253
20	PSNR↑	48.84	50.10	50.19	51.12	50.85	51.56	51.19	51.49	51.56	51.54	51.62	51.71	54.44
	SSIM↑	0.9896	0.9926	0.9927	0.9941	0.9937	0.9946	0.9942	0.9946	0.9946	0.9946	0.9947	0.9948	0.9973
	PSNR-B↑	46.32	48.99	49.17	50.67	50.10	51.09	50.72	50.95	50.93	51.11	51.11	51.12	53.54
	SAM↓	0.0210	0.0178	0.0175	0.0154	0.0162	0.0148	0.0157	0.0151	0.0149	0.0148	0.0147	0.0145	0.0109
	RASE↓	128.806	110.217	108.934	97.196	101.101	93.473	99.126	95.387	94.043	93.812	92.856	91.910	68.196
	ERGAS↓	257.612	220.435	217.867	194.391	202.203	186.947	198.253	190.774	188.087	187.624	185.712	183.821	136.397
30	PSNR↑	50.47	51.63	51.72	52.75	52.40	53.19	52.81	53.14	53.21	53.20	53.26	53.33	55.58
	SSIM↑	0.9927	0.9945	0.9946	0.9959	0.9956	0.9963	0.9959	0.9963	0.9963	0.9963	0.9963	0.9964	0.9979
	PSNR-B↑	48.06	50.75	50.92	52.07	51.57	52.41	52.09	52.32	52.37	52.51	52.52	52.54	54.36
	SAM↓	0.0174	0.0150	0.0147	0.0128	0.0136	0.0123	0.0131	0.0125	0.0123	0.0122	0.0123	0.0121	0.0095
	RASE↓	107.266	92.420	91.377	80.6702	84.7024	77.9083	82.7037	79.3844	78.2077	77.8952	77.2280	76.6624	59.9132
	ERGAS↓	214.532	184.841	182.754	161.340	169.405	155.817	165.407	158.769	156.401	155.791	154.456	153.325	119.826
40	PSNR↑	51.68	52.64	52.73	53.90	53.49	54.35	53.98	54.33	54.39	54.39	54.46	54.50	56.32
	SSIM↑	0.9945	0.9953	0.9953	0.9969	0.9966	0.9971	0.9969	0.9971	0.9972	0.9972	0.9972	0.9972	0.9982
	PSNR-B↑	49.39	51.91	52.08	53.19	52.68	53.32	53.15	53.32	53.31	53.35	53.39	53.42	54.98
	SAM↓	0.0151	0.0134	0.0132	0.0112	0.0120	0.0111	0.0115	0.0109	0.0108	0.0107	0.0106	0.0105	0.0087
	RASE↓	93.6201	82.2431	81.3951	70.7956	74.9112	68.4600	72.6389	69.5384	68.4722	68.1998	67.6413	67.3538	55.1219
	ERGAS↓	187.240	164.485	162.790	141.591	149.823	136.920	145.278	139.077	136.944	136.400	135.283	134.708	110.2437

We conducted comprehensive comparative experiments involving the latest studies to demonstrate the effectiveness of our proposed method. We compared MASCAR, which uses only MNet, with MASCAR+, which uses both MNet and PNet, to other methods. Furthermore, MASCAR+ was designed to assume an efficient data transmission model, with the compression ratio of PAN set higher than that of MS in the data sampling process for training.

We compared the proposed method with other compression artifact removal works on NSIs, such as NSSRC [6], HSSE [7], SwinIR [14], FBCNN [12], COLA-Net [26], Swin2SR [27], ART [28], and GRL [29]. In addition, MIRNetv2 [55] and Restormer [56], which show excellent performance in general image restoration tasks, were also compared. These methods are grouped and referred to as the NSI network group. Furthermore, we compared the proposed method with OTO [48], which was proposed for RSI compression artifact removal. Few recent studies have performed compression artifact removal for RSIs, so we compared the various RSI restoration methods. We included HSID-convolutional neural network (CNN) [30], AODN [34], and NSSNN [33], which were proposed for denoising, MONet [36] for despeckling, and RSDehazeNet [40] and Trinity-Net [42] for dehazing. In addition, RSINet [43] and ARSRN [44] methods proposed for spatial super-resolution and HDRAN [46] and HASIC-Net [47] proposed for spectral super-resolution were also included for comparison. These methods are collectively referred to as the RSI network group.

We reimplemented the networks proposed in each study, and all training parameters were set according to the authors' descriptions in the corresponding papers. For RSINet and ARSRN training, we removed the upsampler at the network's end. For a fair comparison, training QFs were set the same as those of the proposed MASCAR.

C. Synthetic Dataset Experiments

In the synthetic dataset experiments, we configured the subband compression ratios of the input image to be sampled equally. For quantitative performance comparison, we used the peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [57], PSNR including blocking effects (PSNR-B) [58], spectral angular mapping (SAM) [59], relative dimensionless global error in synthesis (ERGAS) [60], and relative average spectral error (RASE) [61] for the quantitative evaluation of restored image. The higher PSNR, SSIM, and PSNR-B values denote the better spatial quality of the restored MS, and the lower SAM, RASE, and ERGAS indicate the better spectral quality of the restored MS. In addition, all metrics were obtained by inference and averaging 143 validation images.

1) NSI Network Group Comparison: Table I presents the quantitative performance comparison results of training and evaluating the NSI network group with the KARI Synthetic

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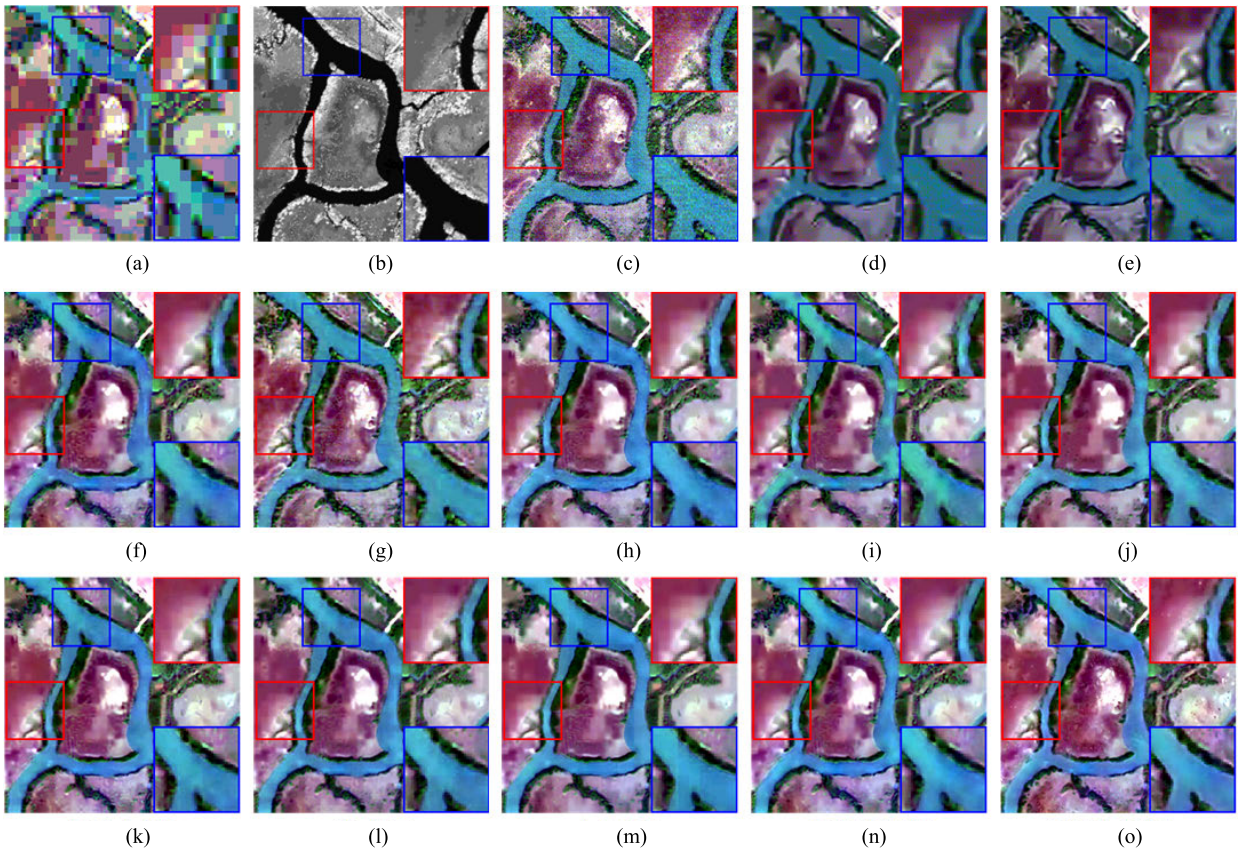


Fig. 9. Compression artifacts removal results of NSI network group on the KARI Synthetic Comp dataset. (a) Compressed MS. (b) Compressed PAN. (c) Ground-truth MS. (d) NSSRC. (e) HSSE. (f) SwinIR. (g) FBCNN. (h) COLA-Net. (i) MIRNetv2. (j) Restormer. (k) Swin2SR. (l) ART. (m) GRL. (n) MASCAR (ours). (o) MASCAR+ (ours).

Comp dataset. These results show that the proposed method outperforms other works. In MASCAR+ evaluation, the QF of PAN was equal to that of MS, and the QFs of PAN used for evaluation were all set to 10. The experimental results indicate that our proposed method attains state-of-the-art performance, as evidenced by metrics such as PSNR, SSIM, PSNR-B, SAM, RASE, and ERGAS. Compared with COLA-Net, a state-of-the-art CNN-based method, the proposed MASCAR demonstrates superior performance across all QFs, with average gains of 0.1475/0.0002/0.0700/0.0004/1.490/2.980 in terms of PSNR, SSIM, PSNR-B, SAM, RASE, and ERGAS, respectively. In addition, compared to the state-of-the-art transformer-based method, Swin2SR, and GRL, MASCAR gained 0.1300/0.0002/0.2000/0.0004/1.976/3.949 and 0.0750/0.0001/0.0675/0.0002/0.7568/1.5130 on average across all QFs, respectively. Furthermore, compared to competing models, our proposed MASCAR+ shows an average gain of at least 2.5700 for PSNR, 0.0026 for SSIM, 2.2525 for PSNR-B, 0.0036 for SAM, 22.818 for RASE, and 45.636 for ERGAS. Fig. 9 shows the compression artifact removal results for the RGB band among the inference results of the QF10 MS image. From the visual quality, model-based methods [6], [7] could hardly remove the compression artifacts, and SwinIR, FBCNN, and MIRNetv2 could not remove most of the blocking artifacts. In addition, Restormer, COLA-Net, Swin2SR, ART, and GRL removed many artifacts but failed to restore fine details, and traces of blocking artifacts remained. In contrast, the proposed MASCAR removed many artifacts and restored details. In addition, MASCAR+ almost completely removed compression artifacts, and the enhanced details of PAN were well fused to show visually pleasing results.

2) RSI Network Group Comparison: Table II lists the results of training and evaluating the RSI network group with the KARI Synthetic Comp dataset. As shown in Table II, the proposed method shows superior quantitative performance than other methods. Compared to the highest quantitative performance results, the proposed MASCAR gained across all QFs, with average gains of 0.3475/0.0002/0.1850/0.0007/2.1745/4.3488 in terms of PSNR, SSIM, PSNR-B, SAM, RASE, and ERGAS, respectively. In addition, MASCAR+ averaged gains of 2.8425/0.0026/2.370/0.0041/24.236/48.471 across all QFs, respectively. Fig. 10 shows the compression artifact removal results for the RGB band among the inference results of the QF10 MS image. Based on the visual quality, HSID-CNN, MONet, RSIDehazeNet, Trinity-Net, and RSINet could hardly remove compression artifacts. In addition, we confirmed that weak blocking artifacts remained in NSSNN, OTO, AODN, and ARSRN. HDRAN and HASIC-Net removed many compression artifacts, but we could still see traces of blocking artifacts. Furthermore, HASIC-Net did not properly restore the boundaries. The proposed MASCAR removed most compression artifacts and provided better visual quality.

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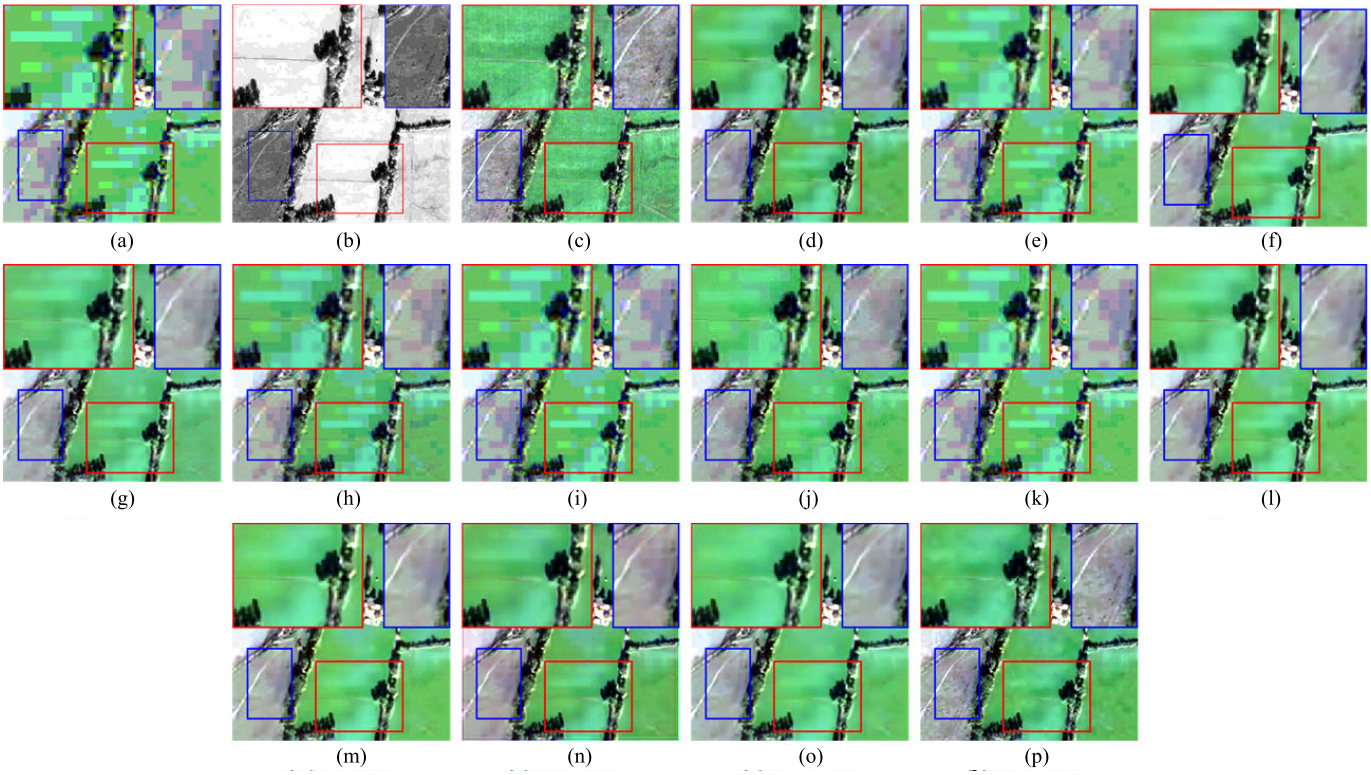


Fig. 10. Compression artifacts removal results of RSI network group on the KARI Synthetic Comp dataset. (a) Compressed MS. (b) Compressed PAN. (c) Ground-truth MS. (d) OTO. (e) HSID-CNN. (f) AODN. (g) NSSNN. (h) MONet. (i) RSDehazeNet. (j) Trinity-Net. (k) RSINet. (l) ARSRN. (m) HDRAN. (n) HASIC-Net. (o) MASCAR (ours). (p) MASCAR+ (ours).

In addition, MASCAR+ completely removed compression artifacts and, at the same time, confirmed that many details were restored.

D. Real-World Dataset Experiments

In the real-world dataset experiments, to model SSVC, we used a channelwise random quality sampling method and configured the subband compression ratio of the input image to be sampled differently. The training was performed on the KARI Synthetic Comp dataset, and the evaluation was done on the KARI Blind Comp dataset. Blind/referenceless image spatial quality evaluator (BRISQUE) [62], naturalness image quality evaluator (NIQE) [63], and perception based image quality evaluator (PIQE) [64] were used as quantitative metrics to evaluate the quality of the images. All three metrics were no-reference quality metrics, and a lower value indicates a better quality image. However, NSSRC [6] and HSSE [7] used the quantization table as a parameter to restore the image. Therefore, we did not perform experiments on a real-world dataset (KARI Blind Comp dataset) with an unknown compression ratio in a remote sensing environment.

1) NSI Network Group Comparison: Table III shows the results of evaluating the NSI network group with the KARI Blind Comp dataset. The proposed method outperforms other works quantitatively. Regarding BRISQUE, NIQE, and PIQE, MASCAR achieves more than 1.890, 0.140, and 3.420 gains, respectively. In addition, MASCAR+ achieves more than 9.360, 0.436, and 13.00 in BRISQUE, NIQE, and PIQE, respectively. Fig. 11 shows the compression artifact removal results of an MS image with SSVC applied with unknown compression ratios. Most NSI network group methods cannot properly remove compression artifacts, with only the FBCNN method alleviating some compression artifacts. As FBCNN is the only method that estimates QF in the NSI network group, it is reasonable to think that it can handle SSVC with unknown compression ratios to some extent. In contrast, the proposed MASCAR successfully removes most compression artifacts and restores edge details. Moreover, the proposed MASCAR+ restores more textures as the details of the PAN are fused.

2) RSI Network Group Comparison: Table IV shows the results of evaluating the RSI network group with the KARI Blind Comp dataset. As shown in Table IV, the proposed method outperforms other works in terms of quantitative metrics. Compared with other works, the proposed MASCAR gains more than 3.19, 0.172, and 3.82 in terms of BRISQUE, NIQE, and PIQE, respectively. In addition, MASCAR+ gains over 10.65, 0.468, and 13.40 in BRISQUE, NIQE, and PIQE, respectively. Fig. 12 illustrates the compression artifact removal results of the MS image with SSVC applied with unknown compression ratios. The HSID-CNN and MONet showed color shifting. Moreover, most of them roughly reconstructed areas with dense edges, but it can be seen that there are many empty parts in the middle of the line, and there is aliasing around the line. The proposed MASCAR restored more lines than other works, and it can be seen that there are fewer empty parts. In addition, MASCAR+ restores more lines and details than MASCAR.

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TABLE II

RSI NETWORK GROUP COMPARISON RESULTS OF AVERAGE QUALITY METRICS ON THE KARI SYNTHETIC COMP DATASET (THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY)

QF	Metric	Methods													
		MS	OTO [48]	HSID-CNN [30]	AODN [34]	NSSNN [33]	MONet [36]	RSDehazeNet [40]	Trinity-Net [42]	RSINet [43]	ARSRN [44]	HDRAN [46]	HASIC-Net [47]	MASCAR (Ours)	MASCAR+ (Ours)
10	PSNR↑	46.51	48.85	48.14	48.99	48.60	47.37	47.79	47.69	47.75	48.77	49.08	49.09	49.34	52.52
	SSIM↑	0.9826	0.9902	0.9986	0.9905	0.9896	0.9875	0.9877	0.9873	0.9873	0.9901	0.9909	0.9910	0.9912	0.9960
	PSNR-B↑	43.95	48.64	47.14	48.92	48.52	46.32	46.54	46.36	46.45	48.55	49.01	48.97	49.21	52.15
	SAM↓	0.0275	0.0203	0.0221	0.2000	0.0209	0.0238	0.0232	0.0235	0.0232	0.0205	0.0196	0.0198	0.0191	0.0136
	RASE↓	168.527	127.833	138.237	125.756	131.502	149.348	143.930	145.482	145.8494	128.670	123.672	123.046	120.678	85.1264
	ERGAS↓	337.055	255.666	276.473	251.513	263.004	298.697	287.859	290.965	291.6987	257.339	247.344	246.092	241.356	170.253
20	PSNR↑	48.84	51.17	50.57	51.32	50.84	49.37	50.18	49.89	50.15	51.06	51.37	51.39	51.71	54.44
	SSIM↑	0.9896	0.9941	0.9934	0.9943	0.9937	0.9923	0.9928	0.9923	0.9926	0.9940	0.9946	0.9946	0.9948	0.9973
	PSNR-B↑	46.32	50.65	49.82	50.86	50.43	48.59	49.23	48.50	49.13	50.47	50.89	50.94	51.12	53.54
	SAM↓	0.0210	0.0155	0.0167	0.0152	0.0162	0.0189	0.0177	0.0183	0.0176	0.0157	0.0150	0.0152	0.0145	0.0109
	RASE↓	128.806	97.8930	104.209	96.1607	101.584	117.921	108.732	112.559	110.236	98.8498	94.8283	94.1028	91.910	68.196
	ERGAS↓	257.612	195.786	208.419	192.322	203.168	235.843	217.464	225.117	220.471	197.670	189.657	188.206	183.821	136.397
30	PSNR↑	50.47	52.79	52.13	52.95	52.43	50.55	51.69	51.35	51.73	52.64	52.91	52.94	53.33	55.58
	SSIM↑	0.9927	0.9959	0.9954	0.9961	0.9956	0.9943	0.9949	0.9945	0.9948	0.9958	0.9962	0.9962	0.9964	0.9979
	PSNR-B↑	48.06	52.05	51.61	52.16	51.73	50.03	51.04	50.07	50.99	51.72	52.15	52.35	52.54	54.36
	SAM↓	0.0174	0.0129	0.0140	0.0127	0.0135	0.0165	0.0149	0.0155	0.0147	0.0131	0.0125	0.0128	0.0121	0.0095
	RASE↓	107.266	81.6160	87.2510	80.1416	84.9976	102.758	91.2981	95.0589	91.9015	82.6980	79.5991	78.8344	76.6624	59.9132
	ERGAS↓	214.532	163.2320	174.5019	160.283	169.995	205.516	182.596	190.118	183.803	165.396	159.198	157.669	153.325	119.826
40	PSNR↑	51.68	53.93	53.13	54.11	53.59	51.26	52.66	52.34	52.79	53.80	54.00	54.07	54.50	56.32
	SSIM↑	0.9945	0.9969	0.9936	0.9970	0.9966	0.9953	0.9959	0.9957	0.9960	0.9968	0.9971	0.9971	0.9972	0.9982
	PSNR-B↑	49.39	53.13	52.73	53.17	52.82	50.89	52.17	51.20	52.23	52.79	53.18	53.29	53.42	54.98
	SAM↓	0.0151	0.0114	0.0126	0.0111	0.0118	0.0153	0.0135	0.0139	0.0130	0.0115	0.0110	0.0113	0.0105	0.0087
	RASE↓	93.6201	71.8369	77.8677	70.4197	74.6615	94.5604	81.6599	84.7030	81.3514	72.6051	70.3335	69.3190	67.3538	55.1219
	ERGAS↓	187.240	143.674	155.735	140.839	149.323	189.121	163.3198	169.406	162.703	145.210	140.667	138.638	134.708	110.2437

TABLE III

NSI NETWORK GROUP COMPARISON RESULTS OF AVERAGE BRISQUE, NIQE, AND PIQE ON THE KARI BLIND COMP DATASET (THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY)

Methods	BRISQUE↓ / NIQE↓ / PIQE↓
Compressed MS	35.29 / 5.496 / 44.70
SwinIR [14]	34.77 / 3.761 / 35.64
FBCNN [12]	29.08 / 3.504 / 36.67
COLA-Net [26]	32.83 / 4.124 / 37.86
MIRNetv2 [55]	41.93 / 3.581 / 38.06
Restormer [56]	35.16 / 4.274 / 36.09
Swin2SR [27]	36.48 / 3.853 / 37.17
ART [28]	30.76 / 3.609 / 30.03
GRL [29]	35.80 / 4.180 / 34.33
MASCAR	27.19 / 3.364 / 26.61
MASCAR+	19.72 / 3.068 / 17.03

TABLE IV

RSI NETWORK GROUP COMPARISON RESULTS OF AVERAGE BRISQUE, NIQE, AND PIQE ON THE KARI BLIND COMP DATASET (THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY)

Methods	BRISQUE↓ / NIQE↓ / PIQE↓
Compressed MS	35.29 / 5.496 / 44.70
OTO [48]	37.82 / 3.802 / 35.64
HSID-CNN [30]	34.80 / 4.379 / 31.51
AODN [34]	41.91 / 3.811 / 35.51
NSSNN [33]	36.99 / 4.120 / 35.72
MONet [36]	38.12 / 4.962 / 40.12
RSDehazenet [40]	43.12 / 3.692 / 36.43
Trinity-Net [42]	31.14 / 3.590 / 32.63
RSINet [43]	37.46 / 3.830 / 39.34
ARSRN [44]	39.12 / 3.536 / 37.62
HDRAN [46]	30.38 / 3.722 / 30.43
HASIC-Net [47]	44.87 / 3.935 / 34.87
MASCAR	27.19 / 3.364 / 26.61
MASCAR+	19.72 / 3.068 / 17.03

E. Analysis of Experimental Results

We investigate the impact of the key component modules of the proposed model and the proposed MDCC on the MDLPCL strategy. All the ablation experiments are performed using the KARI Synthetic Comp dataset.

1) Effectiveness of Each Loss: We studied the effectiveness of each loss function. The experiment was conducted by changing the combination of loss functions that comprise the MDCC. The quantitative scores of the experiment are shown in Table V. Fig. 13 shows the supervised domain outputs, guided by the proposed MDCC. The first column represents the ground truth, the second column denotes the compressed map, and the third to sixth columns are supervised domain outputs for each loss function option. Rows one to five represent the MS output, the MS edge guidance map, the MS mean map, the MS variance map, and the normalized coarse-trained PAN, respectively.

We have the following observations.

1) The MS edge guidance map in Fig. 13. It is apparent that the compressed edge map loses many details and exhibits significant blocking artifacts. When comparing option L_A with other options, it is evident that using

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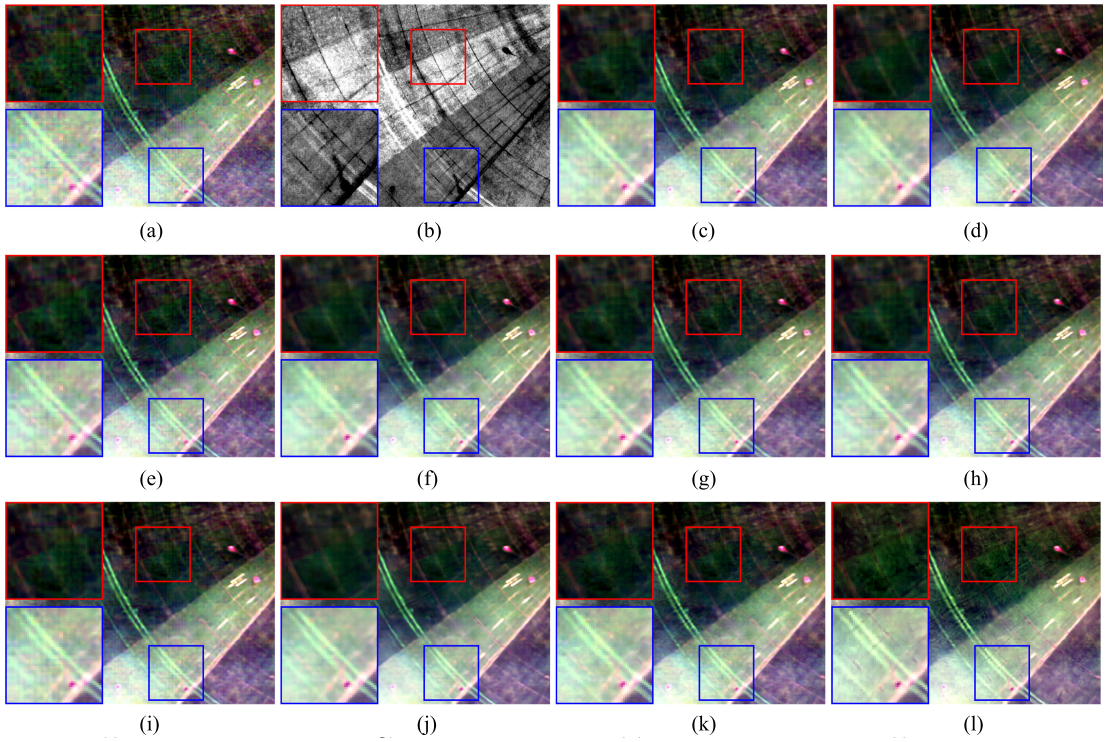


Fig. 11. Compression artifacts removal results of NSI network group on the KARI Blind Comp dataset. (a) Compressed MS. (b) Compressed PAN. (c) SwinIR. (d) FBCNN. (e) COLA-Net. (f) MIRNetv2. (g) Restormer. (h) Swin2SR. (i) ART. (j) GRL. (k) MASCAR (ours). (l) MASCAR+ (ours).

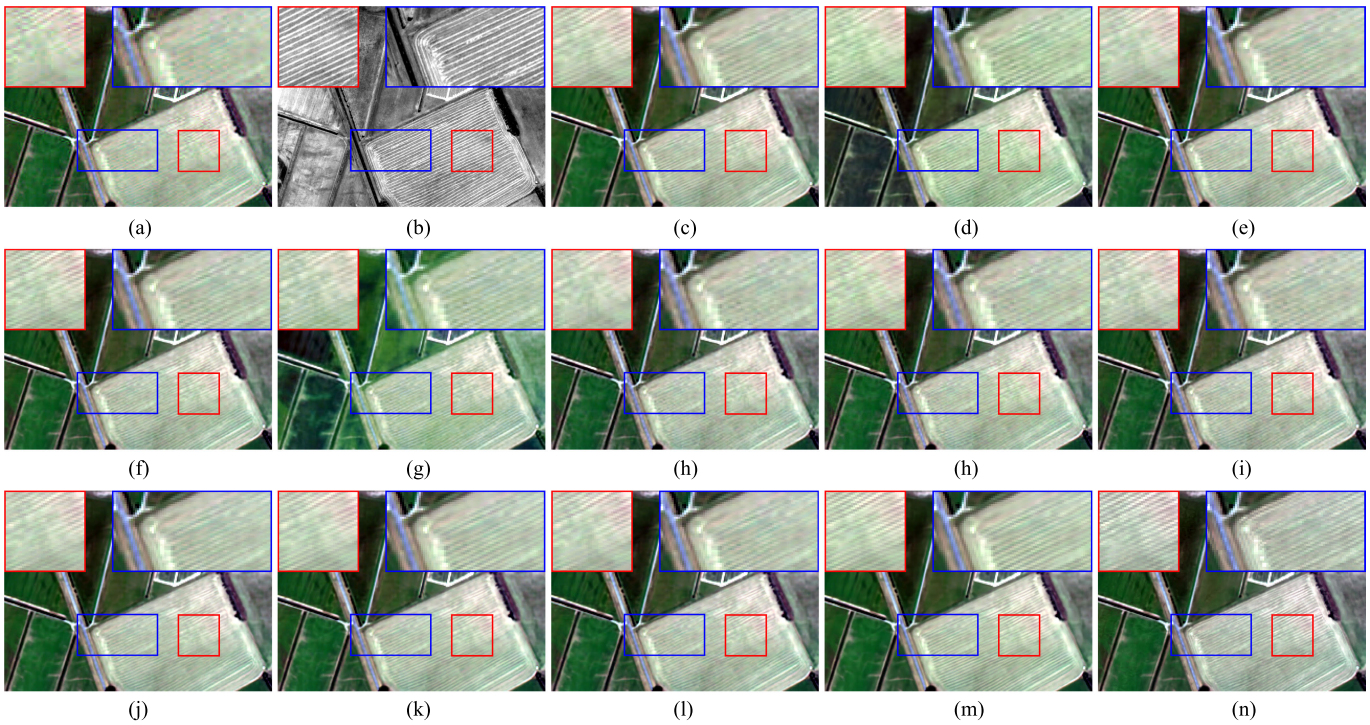


Fig. 12. Compression artifacts removal results of RSI network group on the KARI Blind Comp dataset. (a) Compressed MS. (b) Compressed PAN. (c) OTO. (d) HSID-CNN. (e) AODN. (f) NSSNN. (g) MONet. (h) RSDehazeNet. (i) Trinity-Net. (j) RSINet. (k) ARSRN. (l) HDRAN. (m) HASIC-Net. (n) MASCAR (ours). (o) MASCAR+ (ours).

$\mathcal{L}_E$ and $\mathcal{L}_{\text{EGM}}$, the output of the EDGB restores more details.

2) As shown in Fig. 13, the compressed mean map shows weak blocking artifacts, while the compressed variance map reveals blocking artifacts of medium intensity and color distortion artifacts. When comparing option L_C with option L_A and L_B, it is evident that using $\mathcal{L}_{\text{MV}}$, the output of the MVDB is improved.
3) The MS output in Fig. 13. The option L_D, which uses all the losses constituting MDCC, restores more fine details

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TABLE V
ABLATION EXPERIMENTS ON THE EFFECTIVENESS OF EACH LOSS ON THE PROPOSED MASCAR+

Option	$\mathcal{L}_R$	$\mathcal{L}_{NP}$	$\mathcal{L}_E$	$\mathcal{L}_{EGM}$	$\mathcal{L}_{MV}$	$\mathcal{L}_Q$	PSNR / SSIM / PSNR-B
L_A	✓	✓	✗	✗	✗	✗	52.20 / 0.9959 / 51.79
L_B	✓	✓	✓	✓	✗	✗	52.32 / 0.9959 / 51.92
L_C	✓	✓	✓	✓	✓	✗	52.46 / 0.9960 / 52.13
L_D	✓	✓	✓	✓	✓	✓	52.52 / 0.9960 / 52.15

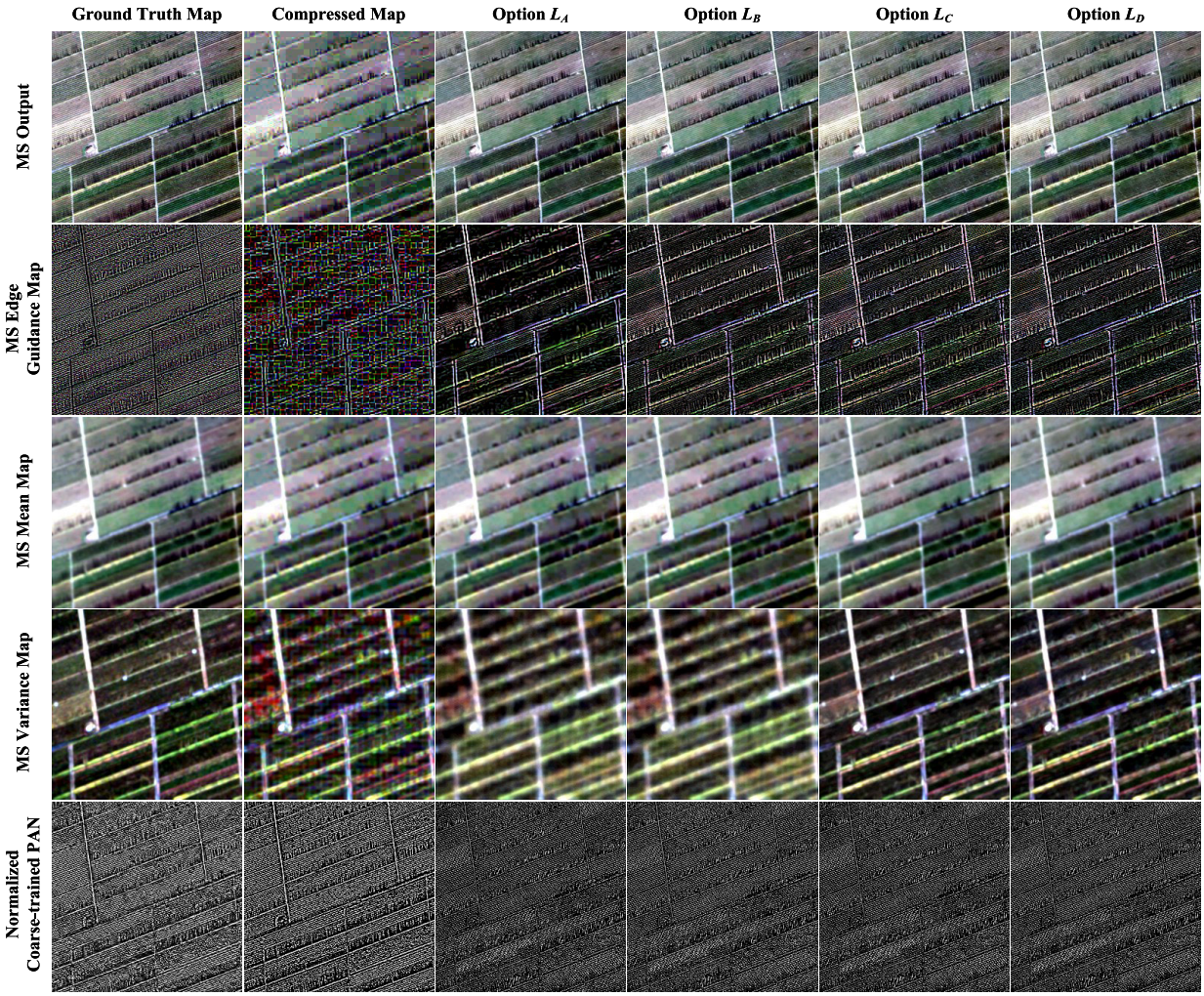


Fig. 13. Perceptual quality comparison of supervised domain outputs from MASCAR+ on the KARI Synthetic Comp dataset. The option L_A uses $\mathcal{L}_R$ and $\mathcal{L}_{\text{NP}}$. The option L_B utilizes $\mathcal{L}_E$ and $\mathcal{L}_{\text{EGM}}$ with option L_A. The option L_C employs $\mathcal{L}_{\text{MV}}$ with L_C. The option L_D incorporates all losses proposed in this article.

than the other results. Furthermore, Table V shows the best performance when all losses are used together. This is because supervised domain outputs are effective priors for subsequent domain training, resulting in good performance. This result demonstrates the effectiveness of the MDCC.

2) Effectiveness of Each Module: We analyzed the effectiveness of the introduced modules proposed in this article. This allows us to examine the effectiveness of the core components of the proposed MASCAR and MASCAR+ modules. The baseline model of the MASCAR consists of an NDB with an FRB. In addition, the baseline model of the MASCAR+ includes PNet with the baseline of MASCAR. The experiments were conducted by incorporating each module into the baseline model. The quantitative scores of the experiment are shown in Table VI. Fig. 14 illustrates the absolute residual maps between the restored MS and the ground truth.

As reported in Table VI, we have the following observations.

1) It was confirmed that the performance using only the MVDB was higher than that using only the EDGB. This is because the MVDB removes compression artifacts reflected in the restored MS output from the denormalization process. When comparing Fig. 14(e) and (f) with Fig. 14(j) and (k), it is observed that the EDGB alone exhibits more errors than the MVDB alone.

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TABLE VI
ABLATION EXPERIMENTS ON THE EFFECTIVENESS OF EACH MODULE

Method	Option	NDB	FRB	MVDB	EDGB	SFQEM-SFQRM, QAM	PSNR / SSIM / PSNR-B
MASCAR	M1	✓	✓	✗	✗	✗	48.61 / 0.9909 / 48.31
	M2	✓	✓	✓	✗	✗	48.95 / 0.9911 / 48.61
	M3	✓	✓	✗	✓	✗	48.86 / 0.9910 / 48.53
	M4	✓	✓	✓	✓	✗	49.01 / 0.9912 / 48.91
	M5	✓	✓	✓	✓	✓	49.34 / 0.9912 / 49.21
MASCAR+	P1	✓	✓	✗	✗	✗	51.74 / 0.9956 / 51.37
	P2	✓	✓	✓	✗	✗	51.99 / 0.9957 / 51.63
	P3	✓	✓	✗	✓	✗	51.88 / 0.9957 / 51.48
	P4	✓	✓	✓	✓	✗	52.26 / 0.9959 / 51.89
	P5	✓	✓	✓	✓	✓	52.52 / 0.9960 / 52.15

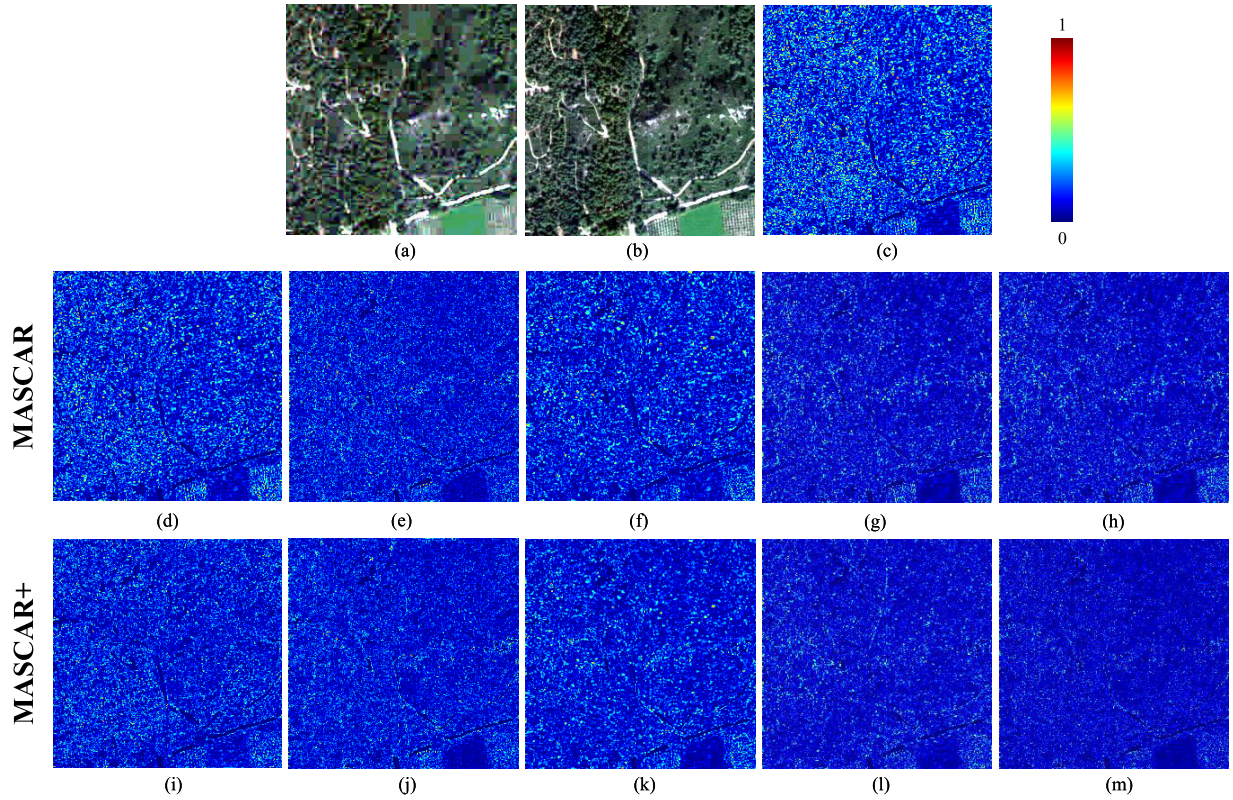


Fig. 14. Perceptual quality comparison of the error maps on the KARI Synthetic Comp dataset. Option M1 is the baseline model of MASCAR, utilizing NRB and FRB. Option P1 is the baseline model of the MASCAR+, incorporating NRB, FRB, and PNet. The options M2 and P2 utilize MVDB with the baseline. The options M3 and P3 employ EDGB with the baseline. Options M4 and P4 use MVDB and EDGB with the baseline. Options M5 and P5 incorporate all modules proposed in this article. (a) Compressed MS. (b) Ground-truth MS. (c) Residual of (a). (d) Residual of option M1. (e) Residual of option M2. (f) Residual of option M3. (g) Residual of option M4. (h) Residual of option M5. (i) Residual of option P1. (j) Residual of option P2. (k) Residual of option P3. (l) Residual of option P4. (m) Residual of option P5.

2) The most substantial quantitative performance improvements were achieved when applying the ICIQA mechanism. This is because the quality of the compressed image was precisely estimated through the SFQEM, and the SFQRM adaptively reconstructed quality features for compression artifact removal. In addition, pixelwise refinement effectively removes the remaining compression artifacts in the QAM. When comparing Fig. 14(g) and (h) with Fig. 14(l) and (m), it is evident that the cases with SFQEM-SFQRM and QAM added show fewer errors.

3) Analysis of Various Compression Artifacts: In Fig. 13, the compressed MS output illustrates the various artifacts, such as blocking, blurring, ringing, aliasing, and color distortion, resulting from lossy compression induced by quantization. To elaborate, in the compressed MS depicted in Fig. 13, blocking artifacts occurred as large as macroblock size, introducing discontinuities in the entire image. Furthermore, blurring artifacts are generated within the macroblocks, resulting in the loss of fine details. In addition, color distortion artifacts are apparent in the image center, blending multiple colors into a single hue. Moreover, ringing and aliasing artifacts occur along the left-hand side of the image's white line border, attributable to the high-frequency loss. The proposed MASCAR successfully removes most compression artifacts, especially blocking artifacts, and restores edge details.

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Specifically, when comparing visual quality with the ground truth, it becomes evident that a significant portion of fine-textured details has been successfully restored.

4) Summary: The proposed method removes more artifacts than other methods because nonredundant enriched features are extracted through the proposed MDLPCL strategy and used for restoration. In addition, supervised domain outputs guided by the MDCC proposed for the MDLPCL strategy serve as effective priors for subsequent domain training, providing excellent visual quality. In addition, our method more effectively restores clear texture details compared to other approaches, a success attributable to the well-conceived design of the DFA and DEA. The proposed MASCAR achieved the best performance on all quantitative metrics for validating compression artifact removal performance on synthetic dataset experiments. Furthermore, MASCAR+ demonstrated a performance improvement of about 2 dB or more in terms of PSNR and PSNR-B compared to other methods. In addition, as a result of checking the visual quality, we confirmed that almost all compression artifacts were removed, and even fine details were restored using the PAN information. In addition, the proposed MASCAR effectively removes compression artifacts from MS, which is applied SSVC with unknown compression ratios in remote sensing environments and achieves the highest performance in the no-reference quality metrics. This result demonstrates the proposed ICIQA mechanism's effectiveness in responding flexibly to SSVC.

V. Conclusion

In this article, we proposed MASCAR, a novel network designed to restore MS images applied to unknown compression ratios in remote sensing environments. We trained our proposed network by applying the MDLPCL strategy and the MDCC. This approach involved decomposing the input image into various spatial domains and then sequentially fusing these using the domain features trained in each domain branch as prior information. In addition, we introduced the DFA to effectively process RSIs, allowing the network to focus on texture detail restoration and the DEA to improve the spatial and edge details in the restored images. Furthermore, the ICIQA mechanism was proposed to flexibly respond to SSVC with unknown compression ratios in remote sensing environments. We have conducted extensive comparative experiments on both synthetic and real-world datasets to demonstrate the superiority of the proposed method over state-of-the-art methods. The results of these experiments clearly indicate that our proposed method surpasses other approaches in terms of both full-reference and no-reference quality metrics and visual quality.

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